08. Exact modular-data and symmetry-anomaly consistency tests

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A proposed quantum theory comes with mathematical rules for combining excitations and transforming its descriptions. These rules should fit together like the pieces of a puzzle. This problem builds exact checks for that fit and then asks a second question: can a chosen global symmetry be made into a gauge symmetry? A theory can be consistent even when this second operation is obstructed, so the two verdicts must remain separate.

Key terms

- **Rational CFT** — A conformal field theory with finitely many basic representation types relative to a specified chiral algebra.
- **Character** — A generating function counting states within one representation of that algebra.
- **Modular S and T matrices** — Matrices describing how the characters change under two basic transformations of the torus.
- **Fusion rules** — Rules specifying which excitation types can occur when two types are combined.
- **Global symmetry** — A transformation applied consistently throughout a system that leaves its laws unchanged.
- **Gauging** — Promoting an appropriate symmetry to a local redundancy and adding the corresponding gauge structure.
- **Anomaly** — An obstruction to implementing a classical symmetry or a proposed gauging consistently in the quantum setting; an 't Hooft anomaly can exist in a valid theory.
- **Cocycle** — Algebraic data satisfying a compatibility identity, used here to encode an anomaly.
- **Bulk and boundary** — The interior gravitational theory and its lower-dimensional boundary description; symmetries can have different interpretations in the two.

Why this matters

Exact consistency checks can detect false candidate data before researchers invest in a full construction. They also clarify which symmetries can be gauged and which require additional structure, such as anomaly cancellation or inflow from another system. This is useful both for studying quantum field theories and for interpreting holographic gravity proposals.

What progress would mean

The first contribution would be a dependable checker with known passing and failing examples. A failure would identify a particular algebraic inconsistency or gauging obstruction, rather than issuing an unsupported verdict about all of quantum gravity.

Core question

For supplied rational-CFT modular data and a specified finite internal symmetry, which algebraic consistency tests can be certified, and which failures obstruct a proposed gauging?

Scope and assumptions

Use a bosonic rational CFT with a declared chiral algebra and finitely many characters. Input exact algebraic S,T matrices, vacuum label, and a nonnegative integer multiplicity matrix M. Start with the Ising model. An optional finite 0-form group G requires separate defect/twisted-sector data and an anomaly cocycle; these cannot be inferred from the untwisted partition function alone.

The result concerns the supplied CFT data and gauging problem. A general bulk quantum-gravity compatibility verdict is outside the baseline.

Mathematical target

Check S†S=I, symmetry of S where assumed, S^2=C, $(ST)^3$=C in the declared T convention, M_{00}=1, MS=SM, MT=TM, and $M_{ij}\in Z\geq 0$. Check Verlinde coefficients Σ_a S_ia S_ja S*_ka/S_0a are nonnegative integers.

For a finite symmetry cocycle $\omega\in Z^3(G,U(1))$, check the cocycle identity exactly. To certify anomaly-free gauging in the stated setting, supply an explicit 2-cochain β with $\delta\beta=\omega$, or prove its nonexistence in a complete cohomology computation. Distinguish this obstruction from inconsistency of an anomalous but valid boundary theory.

Required outputs and proof obligations

Return pass/fail/unresolved for each named axiom, with exact arithmetic traces. Include the number field and embeddings. Even passing every modular-data test gives necessary algebraic consistency only; full CFT sewing and realizability are separate obligations.

Validation and rejection controls

Verify Ising S entries including $1/\sqrt{2}$ in algebraic arithmetic. Reject altered modular relations, noninteger multiplicities, and a failed cocycle identity. Accept the existence of a boundary global symmetry rather than flagging it as automatically inconsistent. Cross-check a small cyclic-group cohomology example.

Milestones and research extension

Baseline: exact modular-data checker on Ising and one second supplied model. Strong: one finite-group twisted-sector/gauging analysis. Research extension: classify a explicitly finite family of data or derive a conditional bulk consequence using a stated holographic theorem.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Harlow and Ooguri, Symmetries in quantum field theory and quantum gravity — Bulk gauge versus boundary global symmetry, including higher-form extensions.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$S^\dagger S = I, \quad S^2 = C, \quad (ST)^3 = C, \quad MS = SM, \quad MT = TM, \quad M_{00} = 1, \quad M_{ij} \in \mathbb{Z}_{\geq 0}.$$

$$N_{ij}^{\ k} = \sum_a \frac{S_{ia} S_{ja} S_{ka}^*}{S_{0a}} \in \mathbb{Z}_{\geq 0}, \qquad \omega \in Z^3(G, U(1)), \quad \delta\beta = \omega.$$

Fix the T convention and any assumed symmetry of S. An explicit cochain β trivializes the specified anomaly; a complete cohomology calculation may prove no such cochain exists. This gauging obstruction does not make an anomalous boundary theory inconsistent, and untwisted modular data alone do not supply defect data.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — boundary and bulk symmetries confused (95–105, 618–620).** Boundary global symmetries can correspond to bulk gauge symmetries. Flagging the Ising global symmetry as a gravity inconsistency reverses the relevant distinction.

2. **Blocking — anomaly logic (101–115).** A nontrivial 't Hooft anomaly can obstruct gauging without making the original theory inconsistent. $\Omega_2\hat{}\text{Spin}=Z_2$ does not itself say that every nontrivial Arf theory has the asserted gravitational anomaly. The rewrite confines the check to explicit symmetry data.
3. **Major — rational versus algebraic arithmetic (88–93, 613–615).** Modular matrices need algebraic numbers, not only rationals. Finite modular matrices also presuppose a suitable rational chiral algebra.
4. **Major — unbounded classification and duplicates.** A central-charge ceiling alone does not define a finite RCFT search. The two 08 descriptions are consolidated, and higher-form/cobordism extensions require their spacetime dimension, symmetry type, and inflow theory before computation.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For supplied rational-CFT modular data and a specified finite internal symmetry, which algebraic consistency tests can be certified, and which failures obstruct a proposed gauging?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact modular-data checker on Ising and one second supplied model. Strong: one finite-group twisted-sector/gauging analysis. Research extension: classify a explicitly finite family of data or derive a conditional bulk consequence using a stated holographic theorem.

Sources supporting the corrections

- Harlow and Ooguri, Symmetries in quantum field theory and quantum gravity — Bulk gauge versus boundary global symmetry, including higher-form extensions.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search,

and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and

resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.